Teaching Statement

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As statistics and data science become increasingly important in science, technology, medicine, and public policy, my central goal as a teacher is to help students develop a rigorous and lasting way of thinking about data, uncertainty, and evidence. I do not want students to experience statistics as a collection of isolated formulas to memorize. Instead, I want them to understand why statistical methods are needed, what assumptions support them, how different ideas in probability and inference fit together, and how quantitative conclusions should be interpreted in real problems.

My teaching philosophy is grounded in three commitments: clarity, rigor, and an inclusive classroom environment. Statistical theory can be abstract, especially when students first encounter likelihood, sufficiency, hypothesis testing, distribution theory, or Bayesian reasoning. I try to make these topics accessible by beginning with the motivation for each idea, then building the mathematical structure carefully, and finally returning to interpretation. When students understand the problem a method is designed to solve, they are much more willing to engage with the technical details that follow.

Since joining Purdue, I have taught both graduate and undergraduate courses that are central to students' training in statistics. My graduate teaching includes STAT 517, *Statistical Inference*, and STAT 528, *Introduction to Mathematical Statistics*, both of which contribute to students' foundational preparation in theoretical statistics. I have also taught STAT 417, *Statistical Theory*, multiple times. Because STAT 417 serves students in statistics, mathematics, computer science, engineering, and related fields, it is especially important to present the mathematical basis of statistical inference in a way that is both precise and welcoming to a broad audience.

In these courses, I emphasize connections among concepts rather than treating each topic as a separate technique. For example, when teaching estimators, likelihood methods, sufficient statistics, and hypothesis tests, I repeatedly ask students to consider what information the data provide, what uncertainty remains, and why a particular procedure is reasonable under a given model. This approach helps students move beyond procedural mastery toward a deeper analytical understanding of statistical inference. It also helps them see that theoretical statistics is not detached from application; it is the language that allows us to reason carefully from data.

Active interaction is essential to this kind of learning. I encourage students to ask questions, test ideas, and learn from mistakes, and I use frequent small questions during lecture to check understanding and lower the barrier to participation. In office hours, I often guide students by first asking where their reasoning became stuck, then helping them reorganize the problem and identify a productive next step. I try not only to answer the immediate question, but also to model general strategies that students can use when they encounter similar problems later.

I also pay close attention to the classroom climate. Students learn difficult material more effectively when they feel respected, when expectations are clear, and when evaluation is fair and consistent. My student evaluations across graduate and undergraduate courses have reflected this priority. In several offerings, including STAT 528 in Spring 2023 and multiple sections of STAT 417 in Fall 2024 and Fall 2025, median evaluation scores reached 5.0 out of 5.0 on most or all reported measures. I am especially encouraged by strong responses related to preparation, fairness, openness to questions, and a welcoming classroom environment, because these are the conditions that allow rigorous learning to take place.

My commitment to teaching also extends beyond the formal classroom. I gave the talk "From Data to Discovery" for Purdue's Undergraduate Women in Science Program, where I aimed to encourage student interest in statistics and data-driven inquiry. Outreach of this kind is important to me because many students first become interested in quantitative fields when they see how statistical thinking connects to discovery, decision-making, and social impact.

Looking ahead, I want my teaching to continue evolving as I gain more experience with different groups of students and different levels of the curriculum. I will continue to refine examples, assignments, and classroom interactions so that students can see both the intellectual beauty and the practical value of statistics. Ultimately, I hope my courses help students leave with technical competence, confidence in their own reasoning, and a durable appreciation for the role of statistics in understanding the world.